

Sustainable Computing

Sally Earl





What is sustainable computing?

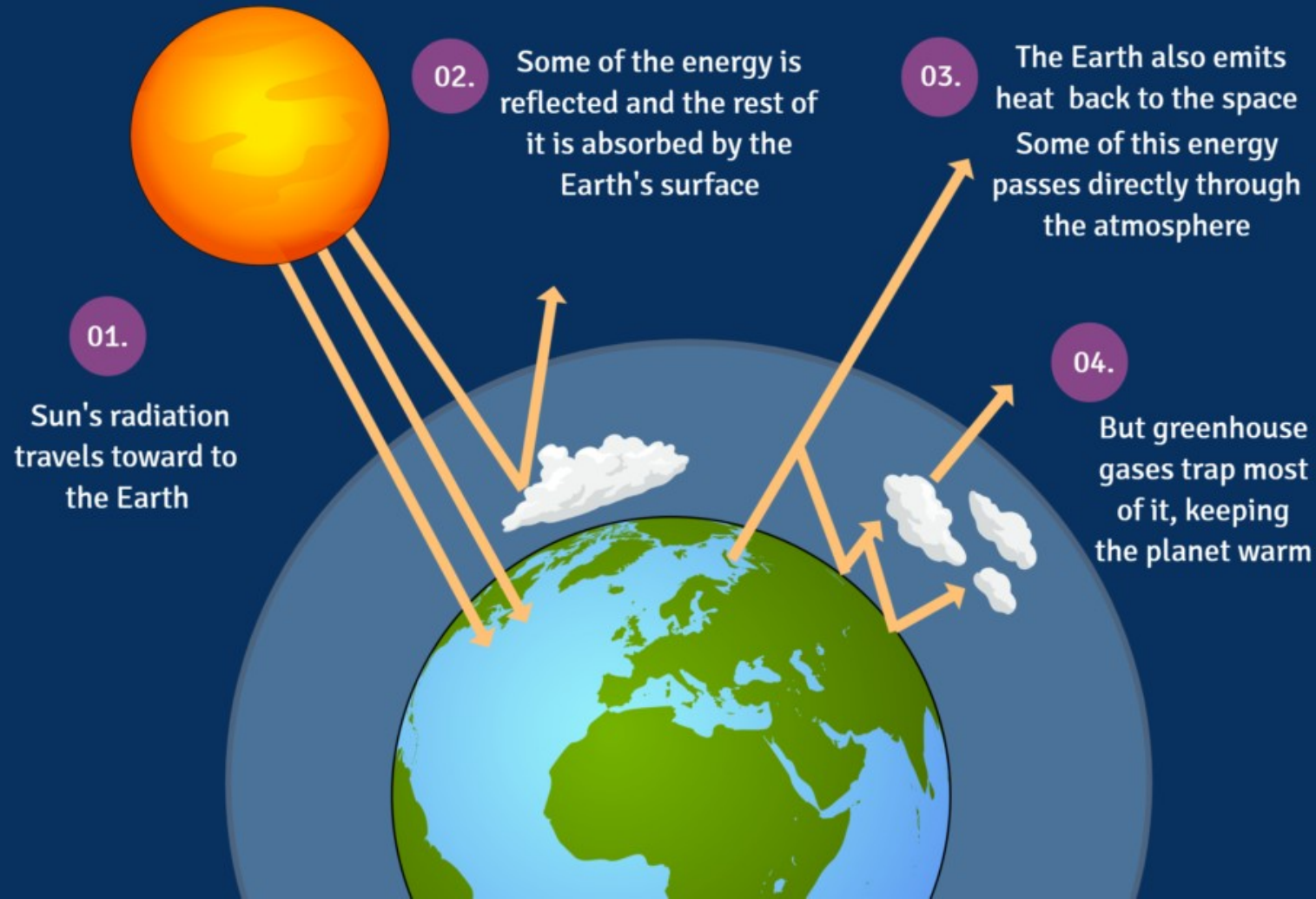
- Sustainable computing involves designing, developing, using, and disposing of computing systems in an environmentally friendly manner



Some Key Terms:

- Greenhouse Gases (GHGs):
 - Gases that trap heat in the atmosphere
 - Main GHGs: Carbon Dioxide, Methane, Nitrous Oxide, Fluorinated Gases for refrigeration
- Global Warming:
 - An increase in Earth's surface temperature caused by GHGs trapping heat
- Climate Change:
 - Long-term changes in Earth's climate, beyond the increase in temperature

The greenhouse effect





Vital Signs

- 2024 was the first year that temperatures averaged 1.5 degrees C above pre-industrial levels
- 2025's maximum arctic ice level was the lowest on record

The Impact of Climate Change

More than 200 killed in Valencia floods as torrential rain hits another Spain region

🕒 1 November 2024

May 2024



November 2024





The Impact of Climate Change

Delhi 'unbearable' as temperatures near 50C

Mount Fuji remains snowless for longer than ever before

Drought leaves Amazon basin rivers at all-time low

The Impact of Climate Change

	1.5° C Increase Millions of Children	1.5° C Increase % of Children	2.7° C Increase Millions of Children	2.7° C Increase % of Children
Heatwaves	62	52	100	83
Crop Failures	23	19	31	26
River Floods	10	8	15	12
Tropical Cyclones	7	6	12	10
Droughts	6	5	8	7

The Impact of Climate Change

	1.5° C Increase Millions of Children	1.5° C Increase % of Children	2.7° C Increase Millions of Children	2.7° C Increase % of Children
Heatwaves	62	52	100	83
Crop Failures	23	19	31	26
River Floods	10	8	15	12
Tropical Cyclones	7	6	12	10
Droughts	6	5	8	7
Wildfires	6	5	10	8



Things to Consider



2 Areas

- There are 2 areas where you should consider sustainability
- The purpose of your solution – is it helping people be sustainable?
- The creation of your solution – is it sustainable to make and/or use?



REDUCE



REUSE



RECYCLE

Reduce: Purpose



- Could your solution help the people of Makers Valley reduce their GHG emissions?

Reduce: Creation



- Could your code be written more efficiently?
- Could you create your system with more environmentally friendly materials?
- Consider the need for using energy-intensive computing such as LLMs

Reuse: Purpose



- Could your solution help people find new purposes for items?
- Could your solution support the makers culture of the area?

Reuse: Creation



- Could your solution be created using discarded or pre-owned materials?

Recycle: Purpose



- Could your solution help people recycle their goods more effectively?

Recycle: Creation



- Could your solution be created using recycled materials?
- Could you ensure the materials your solution is created from can be recycled? What about infinitely recycled?



This list is not exhaustive!

- You do not have to cover any of the things I've mentioned
- But 10% of the marks come from your sustainability considerations
 - Make sure you state them in the demo!



More info on climate change?

- <https://science.nasa.gov/climate-change/>